QEM-8 II: A Hadamard-Tetrahedral-Spinor Framework for Eight-Outcome Qubit Geometry, Coarse-Graining, and History-Sensitive Tests

QEM-8 II：Hadamard—四面体—自旋子框架：八结果量子比特几何、粗粒化与历史敏感检验

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Abstract

This paper develops a mathematically explicit version of the QEM-8 program for a single qubit. The central methodological move is to separate **state geometry**, **operation/history geometry**, and **measurement coarse-graining**, which prevents several distinct eightfold structures from being conflated. The qubit Bloch sphere contains an eight-state cube S_8 encoded by $\mathbb{F}_2^3$. Its even- and odd-parity sectors form a tetrahedral SIC and its antipodal complement, while the three nontrivial translations of the even-parity subgroup form the Klein four group V_4. The corresponding π rotations lift from $SO(3)$ to $SU(2)$ as $-i\sigma_x, -i\sigma_y, -i\sigma_z$, generating the quaternionic sign structure Q_8 and providing a standard-quantum baseline for path-dependent signs.

An eight-outcome cube POVM is then constructed as a joint parent measurement for three orthogonal Pauli observables at equal sharpness $\eta = 1/\sqrt{3}$. Its Walsh-Hadamard decomposition separates one normalization mode, three first-order Bloch modes, three pair-correlation modes, and one parity mode, giving a $1 + 3 + 3 + 1$ decomposition. For the canonical cube POVM, the four higher-order modes vanish identically. A six-outcome Pauli POVM provides a rigorous $8 \to 6 \to 8'$ representation map that preserves the Bloch vector while projecting away the four higher-order Walsh modes. This yields an explicit residual $\Delta = p - p'$ for testing whether an observed eight-outcome distribution contains structure not represented by the standard qubit Bloch sector.

The paper separates **Gate 1A**, in which different operation histories differ only by a standard projective/spinor phase that can be revealed by coherent reference, from **Gate 1B**, a stronger falsifiable hypothesis requiring history-dependent future statistics after the complete standard quantum state/process description and ordinary memory channels have been controlled. No experimental violation of standard quantum mechanics is claimed. The result is a compact mathematical framework,

a set of exact baseline identities, and a hierarchy of tests that can be implemented experimentally or numerically.

中文摘要。本文把 QEM-8 单量子比特方向进一步数学化，并严格区分“状态几何”“操作/历史几何”和“测量粗粒化”。三比特立方体 $\mathbb{F}_2^3$ 给出 8 个 Bloch 球纯态；偶/奇校验各形成一个四面体，偶校验四面体构成 qubit 的 tetrahedral SIC 几何。三个非平凡偶校验平移组成 V_4，其 Bloch 球半转提升到 $SU(2)$ 后形成 Q_8 的符号结构。本文同时构造 8 结果 cube POVM，它是三条正交 Pauli 非锐测量在 $\eta = 1/\sqrt{3}$ 时的联合母测量。利用 H_8 / Walsh-Hadamard 分解，把 8 结果统计拆成 $1 + 3 + 3 + 1$：归一化、3 个 Bloch 一阶模式、3 个二阶相关和 1 个三阶 parity。标准 cube POVM 的后 4 个高阶模式严格为 0。进一步以 6 结果 Pauli POVM 建立严格的 $8 \to 6 \to 8'$ 表示投影，只保留 3 个 Bloch 模式并删除 4 个高阶模式，从而定义残差 $\Delta = p - p'$。本文把标准量子可解释的历史相位称为 Gate 1A，把排除标准量子状态、过程、环境记忆和仪器记忆后仍存在的历史依赖称为强 Gate 1B。本文不声称已经发现超出标准量子力学的新物理，而是给出一套可验证、可否证的数学与实验框架。

Keywords / 关键词: QEM-8; qubit; SIC-POVM; Walsh-Hadamard transform; Clifford group; quaternion group; joint measurability; coarse-graining; holonomy; quantum history / 量子比特；SIC-POVM；Walsh-Hadamard；Clifford 群；四元数群；联合可测性；粗粒化；历史效应。

1. Scope and claim boundary

QEM-8 was originally motivated by direction-sensitive collapse patterns and the possibility that different histories may become indistinguishable under a coarse observation while remaining distinguishable at a deeper process level. The present paper does not assume that such a deeper level is physically real. Instead, it asks a narrower question: **what is the smallest mathematically explicit qubit framework in which eightfold state geometry, four-state tetrahedral structure, two-outcome coarse-graining, path-dependent operation signs, and falsifiable history tests can be placed in one common language?**

A central distinction is required from the outset. There are at least three inequivalent “eights”:

1. **State cube S_8:** eight distinct Bloch-sphere pure states at the vertices of a cube. Antipodal points are orthogonal physical states.
2. **Spinor sign double cover:** for a given Bloch ray, $|\psi\rangle$ and $-|\psi\rangle$ describe the same isolated physical state. The sign becomes meaningful only relative to a coherent reference.
3. **Operation set $O_8 \cong Q_8$:** $\{\pm I, \pm U_x, \pm U_y, \pm U_z\}$, where opposite signs induce the same $SO(3)$ rotation but differ in $SU(2)$.

Conflating these three structures would turn a useful model into an overstatement. The contribution of this paper is therefore primarily a **synthesis and test architecture**, not a claim that standard identities such as SIC geometry, the $SU(2) \to SO(3)$ double cover, or Pauli joint measurability are newly discovered.

2. Three-bit cube and the Hamming-Bloch map

Let

$$s = (s_1, s_2, s_3) \in \mathbb{F}_2^3,$$

and define the Bloch vector

$$\mathbf{r}(s) = \frac{1}{\sqrt{3}}((-1)^{s_1}, (-1)^{s_2}, (-1)^{s_3}).$$

This gives eight unit vectors and therefore eight pure qubit states. For two labels s, t, let $d_H(s,t)$ be the Hamming distance. Directly,

$$\mathbf{r}(s) \cdot \mathbf{r}(t) = 1 - \frac{2d_H(s,t)}{3}.$$

For pure qubit states the fidelity is

$$F(s,t) = \frac{1 + \mathbf{r}(s) \cdot \mathbf{r}(t)}{2},$$

hence

$$\boxed{F(s,t) = 1 - \frac{d_H(s,t)}{3}}. \tag{1}$$

Thus the four Hamming shells $d_H = 0, 1, 2, 3$ map exactly to fidelities

$$1, \quad \frac{2}{3}, \quad \frac{1}{3}, \quad 0,$$

with multiplicities $1 : 3 : 3 : 1$ around every cube vertex.

Proposition 1: parity-tetrahedral decomposition

The even-parity subset

$$H = \{000, 011, 101, 110\}$$

is a two-dimensional linear subspace of $\mathbb{F}_2^3$. Any two distinct members have Hamming distance two, hence by Eq. (1) pairwise fidelity $1/3$. These four states therefore form the standard qubit tetrahedral SIC geometry. The odd-parity coset

$$H + 111 = \{111, 100, 010, 001\}$$

is the antipodal tetrahedron. Each s is orthogonal to $s \oplus 111$.

This yields the exact decomposition

$$\boxed{\mathbb{F}_2^3 = H \sqcup (H + 111)}. \tag{2}$$

The state cube can therefore be viewed as **SIC + anti-SIC**, while remaining an eight-state set of physically distinct Bloch directions.

3. The V_4 layer and Hadamard tomography

The three nonzero elements of H are

$$X = 011, \qquad Y = 101, \qquad Z = 110.$$

Under XOR translation they satisfy

$$X^2 = Y^2 = Z^2 = 0, \qquad X \oplus Y = Z,$$

with cyclic permutations. Hence

$$H \cong V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

On the tetrahedral labels $A = 000, B = 011, C = 101, D = 110$, these translations act as

$$X = (AB)(CD), \qquad Y = (AC)(BD), \qquad Z = (AD)(BC). \tag{3}$$

They are exactly the three nontrivial $2 + 2$ partitions of four objects.

Now take the tetrahedral SIC POVM effects

$$E_j = \frac{1}{4}\left(I + \mathbf{r}_j \cdot \sigma\right), \qquad j \in \{A, B, C, D\}.$$

For a qubit state

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \sigma),$$

let $p = (p_A, p_B, p_C, p_D)^T$ with $p_j = \mathrm{Tr}(\rho E_j)$. Define

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Then

$$\boxed{H_4 p = \begin{pmatrix} 1 \\ r_x/\sqrt{3} \\ r_y/\sqrt{3} \\ r_z/\sqrt{3} \end{pmatrix}}. \tag{4}$$

Thus the same H_4 that is the character table of V_4 is also a direct tomography transform between four SIC probabilities and three Bloch coordinates.

The three $2+2$ coarse-grainings are

$$\begin{aligned} E_A + E_B &= \frac{1}{2}\left(I + \frac{\sigma_x}{\sqrt{3}}\right), \\ E_A + E_C &= \frac{1}{2}\left(I + \frac{\sigma_y}{\sqrt{3}}\right), \\ E_A + E_D &= \frac{1}{2}\left(I + \frac{\sigma_z}{\sqrt{3}}\right). \end{aligned} \tag{5}$$

They are therefore binary unsharp Pauli measurements with sharpness $1/\sqrt{3}$.

4. From V_4 to the quaternionic spinor lift

Place the tetrahedral Bloch vectors at

$$A = \frac{(1,1,1)}{\sqrt{3}}, \quad B = \frac{(1,-1,-1)}{\sqrt{3}}, \quad C = \frac{(-1,1,-1)}{\sqrt{3}}, \quad D = \frac{(-1,-1,1)}{\sqrt{3}}.$$

A π rotation around the Cartesian x, y, z axes induces precisely the permutations in Eq. (3). At the $SO(3)$ level these three half-turns commute and square to identity, reproducing V_4.

For spin-$1/2$, choose the $SU(2)$ lifts

$$U_x = -i\sigma_x, \qquad U_y = -i\sigma_y, \qquad U_z = -i\sigma_z.$$

Then

$$U_x^2 = U_y^2 = U_z^2 = -I, \tag{6}$$

and, with this sign convention,

$$U_x U_y = U_z, \qquad U_y U_x = -U_z. \tag{7}$$

Therefore

$$\{\pm I, \pm U_x, \pm U_y, \pm U_z\} \cong Q_8, \tag{8}$$

while quotienting by the center $\{\pm I\}$ recovers V_4.

Equation (7) is a precise standard-quantum example of "same visible endpoint, different lifted history." On the Bloch sphere the two histories XY and YX produce the same π rotation. At the spinor-operation level they differ by the central sign $-I$.

Gate 1A: standard projective-history baseline

The difference between U_z and $-U_z$ is invisible when either acts on an isolated density matrix. It becomes observable only after embedding the alternatives into coherent branches so that the sign becomes a relative phase. For example, a control qubit can coherently select $U_x U_y$ or $U_y U_x$, converting the hidden sign into a measurable phase on the control. This is standard quantum mechanics and is not evidence for new hidden physics.

5. The eight-outcome cube POVM and exact $8 \to 4 \to 2$ marginalization

For $s = (s_x, s_y, s_z)$ with each $s_i \in \{\pm 1\}$, define

$$\boxed{G_s = \frac{1}{8}\left[I + \frac{s_x\sigma_x + s_y\sigma_y + s_z\sigma_z}{\sqrt{3}}\right].} \tag{9}$$

The eight G_s are positive and sum to I. Summing over two signs gives

$$\sum_{s_y, s_z} G_s = \frac{1}{2}\left(I + \frac{s_x\sigma_x}{\sqrt{3}}\right), \tag{10}$$

with analogous expressions for y, z. Hence Eq. (9) is a joint parent POVM for three orthogonal Pauli observables at equal sharpness

$$\boxed{\eta = \frac{1}{\sqrt{3}}}. \tag{11}$$

This is the standard triplewise joint-measurability boundary for three orthogonal unbiased qubit observables.

The outcome labels themselves realize a strict marginal hierarchy:

$$(s_x, s_y, s_z) : 8 \longrightarrow (s_x, s_y) : 4 \longrightarrow s_x : 2. \tag{12}$$

This is a mathematically and operationally precise realization of "full three-coordinate outcome $\rightarrow$ lower-dimensional projection."

Three different notions of $8 \rightarrow 4$ must be distinguished:

- **M-type (marginalization):** forget one coordinate.
- **P-type (parity selection):** select one of the two four-point tetrahedral parity sectors.
- **Q-type (antipodal quotient):** identify s and $-s$.

They have the same count but different mathematical and physical meanings.

6. Walsh-Hadamard decomposition: $1 + 3 + 3 + 1$

For an arbitrary probability distribution $p(x, y, z)$ on $x, y, z \in \{\pm 1\}$, the complete Walsh expansion is

$$\boxed{p(x,y,z) = \frac{1}{8}\left[1 + m_x x + m_y y + m_z z + c_{xy} xy + c_{yz} yz + c_{zx} zx + \tau xyz\right].} \tag{13}$$

The seven nontrivial coefficients split naturally into

$$\boxed{7 = 3 + 4}, \tag{14}$$

where

$$m = (m_x, m_y, m_z)$$

contains the three first-order marginals and

$$h = (c_{xy}, c_{yz}, c_{zx}, \tau)$$

contains three pair correlations and one triple parity.

For the canonical cube POVM of Eq. (9), a qubit state with Bloch vector **r** gives

$$p_s = \frac{1}{8}\left[1 + \frac{s_x r_x + s_y r_y + s_z r_z}{\sqrt{3}}\right]. \tag{15}$$

Therefore

$$m_i = \frac{r_i}{\sqrt{3}}, \qquad c_{xy} = c_{yz} = c_{zx} = \tau = 0. \tag{16}$$

In other words, the ideal fixed cube POVM occupies only the **degree-0 + degree-1 Walsh sector**.

This gives a useful conceptual boundary: four higher-order coefficients are mathematically available in a generic three-bit joint distribution, but for a fixed ideal cube POVM on a single qubit they are not independent physical state coordinates.

7. Same marginals, different joint parents

The tetrahedral SIC and anti-SIC provide an instructive standard-quantum baseline for "same visible marginals, different joint structure." Label the tetrahedral outcomes by the even-parity signs

$$(+++), (+--), (-+-), (--+),$$

so that $xyz = +1$. The anti-tetrahedral outcomes have $xyz = -1$. Both four-outcome POVMs have exactly the same three binary marginals in Eq. (5), even though their higher-order joint correlations differ.

A one-parameter family can be formed by choosing the tetrahedral parent with probability λ and the anti-tetrahedral parent with probability $1 - \lambda$. Let

$$\delta = 2\lambda - 1.$$

Then the single-axis means are independent of λ,

$$\mathbb{E}[x] = \frac{r_x}{\sqrt{3}}, \quad \mathbb{E}[y] = \frac{r_y}{\sqrt{3}}, \quad \mathbb{E}[z] = \frac{r_z}{\sqrt{3}}, \tag{17}$$

while

$$\mathbb{E}[xy] = \delta\frac{r_z}{\sqrt{3}}, \quad \mathbb{E}[yz] = \delta\frac{r_x}{\sqrt{3}}, \quad \mathbb{E}[zx] = \delta\frac{r_y}{\sqrt{3}}, \tag{18}$$

and

$$\mathbb{E}[xyz] = \delta. \tag{19}$$

Thus identical first-order marginals do not uniquely identify the parent joint measurement. This is a standard measurement-theoretic phenomenon and provides a baseline against which any stronger QEM-8 history claim must be tested.

8. A rigorous $8 \to 6 \to 8'$ representation projection

Define the six-outcome Pauli POVM

$$F_{i,\pm} = \frac{1}{6}(I \pm \sigma_i), \qquad i \in \{x, y, z\}. \tag{20}$$

For a state with Bloch vector $\mathbf{r}$,

$$q_{i,\pm} = \frac{1}{6}(1 \pm r_i), \qquad r_i = 3(q_{i,+} - q_{i,-}). \tag{21}$$

The canonical cube POVM contains the same Bloch vector via

$$r_i = \sqrt{3}\, m_i. \tag{22}$$

Therefore the following representation map is exact whenever the extracted m is qubit-compatible, i.e. $|\sqrt{3}m| \leq 1$:

$$p_8 \to \mathbf{r} \to q_6 \to \mathbf{r} \to p'_8, \tag{23}$$

with

$$p'_s = \frac{1}{8}\left[1 + \frac{s_x r_x + s_y r_y + s_z r_z}{\sqrt{3}}\right]. \tag{24}$$

In Walsh coordinates this is a projection

$$\boxed{(1+3+3+1) \longrightarrow (1+3)}. \tag{25}$$

The four higher-order modes are erased:

$$c'_{xy} = c'_{yz} = c'_{zx} = \tau' = 0. \tag{26}$$

Define the residual

$$\boxed{\Delta = p - p'.} \tag{27}$$

Then Δ lies entirely in the four-dimensional high-order Walsh sector. If p was generated by the ideal fixed cube POVM on a qubit, $\Delta = 0$ exactly. If a nonzero residual is observed experimentally, it does **not** immediately imply hidden physics; calibration error, measurement-context dependence, extra Hilbert-space dimensions, instrument memory, and system-environment correlations must be tested first.

This projection gives a precise interpretation of the notation $8 \to 6 \to 8'$: the intermediate six-result representation preserves the standard Bloch information, while $8'$ is the canonical eight-result distribution reconstructed from that Bloch information alone. A prime denotes the possible loss of higher-order joint structure.

9. Gate hierarchy and falsifiability

The framework suggests a hierarchy of increasingly strong tests.

Gate 0: algebraic baseline

Verify the exact identities in Eqs. (1)-(26), including SIC geometry, V_4 action, Q_8 spinor lift, cube POVM marginals, and Walsh-sector null relations. These are standard-mathematical/standard-quantum baselines.

Gate 1A: coherent history phase

Prepare two coherent operation histories such as $U_x U_y$ and $U_y U_x$. They have the same action on the isolated qubit density matrix but differ by a relative sign in $SU(2)$. A coherent control or interferometric reference can reveal this sign. Successful observation validates path-sensitive phase bookkeeping, but remains within standard quantum theory.

Gate 1B: strong history-residual test

The stronger hypothesis is:

> Two preparations or histories are operationally indistinguishable under the complete calibrated standard qubit state/process description at time t, yet after the same later intervention they produce reproducibly different future statistics.

Operationally, one may represent a candidate extension as

$$w = (m; h), \qquad m \in \mathbb{R}^3, \quad h \in \mathbb{R}^4,$$

with a linearized future map

$$m' = Am + Ch, \qquad h' = Dm + Bh. \tag{28}$$

The standard isolated-qubit baseline contains no independently physical h sector accessible by a fixed memoryless POVM; equivalently, the observed statistics are determined by ρ. A strong QEM-8 residual would require reproducible evidence that histories with the same calibrated standard description produce different effective h and that a later analyzer converts this into $Ch \neq 0$.

Before interpreting such a residual as new physics, experiments must exclude at least:

1. POVM miscalibration or context dependence;

2. leakage outside the intended two-level Hilbert space;
3. system-environment correlations not captured by reduced-state tomography;
4. instrument or control memory;
5. ordinary non-Markovian dynamics;
6. coherent reference paths carrying standard relative phase;
7. preparation drift or selection bias.

Failure to survive these controls falsifies the strong interpretation while leaving the mathematical QEM-8 representation intact.

10. Symmetry closure and the Clifford connection

The three V_4 half-turns may be combined with all permutations of the three Cartesian axes. Since

$$V_4 \rtimes S_3 \cong S_4, \tag{29}$$

this generates the 24 orientation-preserving symmetries of the cube/octahedron. The projective single-qubit Clifford group is isomorphic to this same S_4, while the Pauli subgroup modulo phases is V_4. Thus the QEM-8 "three absolute-opposition operations + axis permutation" language embeds naturally into the standard single-qubit Clifford symmetry hierarchy.

At the spinor level the corresponding double-cover chain is

$$Q_8 \lhd 2T \lhd 2O, \tag{30}$$

which projects to

$$V_4 \lhd A_4 \lhd S_4. \tag{31}$$

These facts again serve as baseline structure, not as evidence of physics beyond the standard theory.

11. What is new here, and what is not

The following ingredients are standard or direct consequences of standard theory: qubit tetrahedral SIC geometry; the $SU(2)$ double cover of $SO(3)$; Pauli/Clifford symmetries; unsharp joint measurement of Pauli observables; and Walsh-Hadamard analysis of three binary variables.

The proposed contribution is the **specific QEM-8 synthesis**:

- three inequivalent eightfold structures are separated rather than conflated;
- the binary cube, parity tetrahedra, V_4, H_4, spinor Q_8, and joint Pauli measurement are placed into one explicit chain;
- the complete eight-outcome statistics are organized as $1+3+3+1$ Walsh modes;

- the $8 \to 6 \to 8'$ map is defined as a Bloch-information projection with an exact high-order residual;
- Gate 1A is explicitly separated from a stronger Gate 1B, preventing known projective-phase phenomena from being misidentified as new hidden-history physics.

This framework is therefore suitable for a preprint as a mathematical/experimental proposal. Its stronger physical interpretation remains a hypothesis until Gate 1B survives standard controls.

12. Conclusion

The QEM-8 program can be expressed without relying on visual analogy alone. An eight-state Bloch cube has an exact three-bit encoding. Its parity sectors are tetrahedral SIC/anti-SIC structures; its even-parity translations form V_4; the corresponding spin-$1/2$ half-turn operations lift to a Q_8 sign structure; and an eight-outcome cube POVM jointly realizes three orthogonal unsharp Pauli observables at the $1/\sqrt{3}$ boundary. Walsh-Hadamard coordinates separate the standard three-dimensional Bloch sector from four higher-order joint-correlation modes. A Pauli-six representation then gives an exact $8 \to 6 \to 8'$ projection that removes those higher-order modes.

The main methodological conclusion is that **history should first be modeled as structure of paths, operations, phases, or joint couplings rather than automatically as additional static states**. The standard quantum theory already contains nontrivial path lifts and projective signs. The scientifically meaningful frontier is therefore a strong residual test: whether history can influence future statistics after the full standard state/process description and ordinary memory channels have been controlled. The framework presented here turns that question into a sequence of explicit, falsifiable gates.

Reproducibility note

An accompanying Python script, `qem8_verify.py`, checks the central finite-dimensional identities used in this manuscript: the Hamming-fidelity formula, tetrahedral SIC overlap, V_4 XOR algebra, Q_8 spinor signs, H_4 tomography transform, cube POVM marginals, the H_8 high-order null sector, and the canonical $8 \to 6 \to 8'$ reconstruction.

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本稿建议以“数学框架 + 标准量子嵌入 + 强可证伪 Gate”定位发布。文中所有 SIC、Bloch、SU(2)/SO(3)、Clifford、联合非锐测量等标准结果都明确作为已知理论基线；QEM-8 的新内容放在“如何把这些结构统一成一套历史敏感检验框架”和“如何设计 Gate 1B”上。河图、洛书、易经等历史图式没有放入主证据链，以避免把观察灵感与现代物理证据混在一起。它们可另写一篇“结构启发与跨文化观察模型”的独立文章或作为后续附录。